

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Philadelphia, PA
Expected May 2028
Expected May 2027

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Made the company's AI agent usable by studios outside it** — a typed **SDK** with versioned APIs, per-session isolation and capability enforcement, **1,301 tests**, published over a hosted multi-tenant service and proven inside a game mode an **external builder authored** — then tuned its in-world behaviour so it stopped provoking encounters too hard for children
- **Redirected the team's roadmap** with a **Python multi-agent tool on the Claude Agent SDK** that mined production data through read-only MCP — it showed a reported retention gain was a measurement artifact rather than a real effect, and surfaced the churn drivers that actually mattered
- **Let a builder hold a spoken conversation with the in-world character from a browser**, no game client — hosted speech-to-text and text-to-speech end to end, plus a Claude skill that authors and validates a new mode's config
- **Built the company's lifecycle notification system, then made it autonomous** — it proposes sends, clears its own safety guardrails and delivers unattended at **91% device reach**, producing its first paid conversion

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet app-store requirements and sustain availability

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the **13-shader GLSL pipeline** I owned: PCF shadow mapping, screen-space reflections, vertex AO, a day-night cycle and post-process water through a framebuffer pass

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops in a Qt editor

Capsule | React, Three.js (R3F), Blender

Jan 2025 – Apr 2025

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement
- **Owned how the product felt to use** across the create → contribute → unlock flow, over a Node/MongoDB API

TECHNICAL SKILLS

AI & Agents: Claude Agent SDK, MCP servers, Claude Code, Codex, LLM APIs (Anthropic/Gemini), agent orchestration

Languages: Python, TypeScript, JavaScript, Java, Kotlin, C++, C#, SQL

Frameworks & Infrastructure: React, Next.js, Node.js, React Native (Expo), PostgreSQL/Supabase, Deno, Docker, GCP, AWS, MongoDB, Jest, PostHog, Sentry, Figma

Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing